

Define Project:

The goal of this project is to build a system that predicts how popular a song will be on Spotify using audio features and other information from the Spotify Web API. The main idea is to figure out which musical qualities of a song, such as energy, danceability, or tempo, have the biggest influence on its popularity score. Spotify's popularity metric comes from listener behavior and recommendation patterns, but it is not always clear why certain songs perform better than others. By combining data management, machine learning, and data exploration, this project aims both to predict popularity and better understand the kinds of songs that tend to attract more listeners.

This project connects directly to the topics covered in class because it uses several important data management steps. The first step is collecting data through the API and storing it in a structured relational database. Creating tables for tracks, artists, and audio features while using constraints, foreign keys, and indexing helps maintain clean and organized data. This reflects SQL design principles learned in class and demonstrates why thoughtful schema design is important when working with large or growing datasets.

After storing the data, the next step is to clean and prepare it so it can be used for analysis and modeling. This involves checking for missing values, fixing formatting problems, and turning different features into forms that are consistent and easy to work with. These steps connect to the importance of reliable data cleaning, transforming data into meaningful structures, and ensuring information is organized before building machine learning models.

The machine learning portion of the project uses regression models to predict Spotify popularity scores. Models such as linear regression and random forest regression are trained and compared using standard evaluation methods that measure how close predictions are to the actual popularity values. In addition to predicting the scores, the project also examines which audio features influence popularity the most. This demonstrates not only how to evaluate model quality, but also how to interpret what the model is emphasizing.

Overall, this project brings together the major concepts from the course by showing how good data collection, database design, and data cleaning lead to better analysis and predictions. It highlights the importance of strong data management practices in machine learning workflows and demonstrates how an organized pipeline can lead to meaningful insights about real-world data.

Novelty and Importance

Predicting how popular a song will become before or after its release has become a major challenge for artists, producers, and music platforms. Even though Spotify provides a large amount of metadata and audio feature information, there is still no direct way to fully understand what makes a track successful. The goal of this project is to investigate that gap by building a model that predicts Spotify track popularity using measurable audio traits and artist-level

information. What makes this project interesting is that it combines data engineering, machine learning, and real-world music industry questions in a way that feels both practical and creative.

One reason this project is important is because music consumption patterns are increasingly shaped by algorithms. Playlists such as “Discover Weekly” and “Release Radar” influence which artists gain exposure, while popularity metrics affect how songs are recommended and categorized. Spotify’s popularity measurement is somewhat opaque because it combines total plays, recency, and engagement patterns in a way that is not publicly disclosed. By modeling popularity using available features, this project attempts to uncover whether certain musical characteristics relate to listener behavior in meaningful ways. Understanding these relationships can provide insight into both music analytics and how digital platforms shape cultural trends.

Another important aspect of the project is the way the data is collected and organized. Instead of relying entirely on a pre-cleaned dataset, this project utilizes track, artist, and feature data directly from the Spotify API. The information is stored inside an organized SQL database, which serves as the foundation for modeling and analysis. This creates a workflow that reflects real-world data management practices used in industry settings. Working with live API data also ensures the project includes more recent music trends instead of relying only on outdated datasets.

Many music analytics studies focus primarily on genre classification or recommendation systems. Fewer projects examine predicting Spotify popularity directly, especially when combining connected data across tracks, artists, albums, and audio features. By integrating these sources and studying how they interact, this project provides a more complete understanding of what “popular” means on modern music platforms.

The motivation for this project comes from both curiosity and personal interest in Spotify and music analytics. Music is an area where many people believe they can predict whether a song will become popular after hearing only a few seconds. This project attempts to test those assumptions using data-driven analysis. Additionally, the technical workflow involving API extraction, database design, feature engineering, and predictive modeling strongly connects with concepts learned throughout the course.

Plan

For this project, data was collected from the Spotify API. The dataset includes variables such as track-level information (name, ID, release date), artist-level characteristics (artist popularity and genres), and Spotify metadata. Information related to current music trends was gathered by pulling songs from top charts and curated playlists using authenticated Spotify API calls in Python.

After collecting the data, it was organized into a relational SQL database with multiple tables including Tracks, Artists, and AudioFeatures. Foreign keys, primary keys, and indexes were implemented to maintain structure and organization. The data cleaning process included removing duplicates, handling null values, and standardizing formatting. Additional features

were engineered to improve the machine learning model, including release year and release month variables.

In terms of modeling, both basic regression methods and machine learning algorithms were used. Linear regression was first applied to identify which variables were most related to Spotify popularity. More advanced models such as Random Forest Regression were then used to capture more detailed relationships between features. The models were built in Python using scikit-learn.

The implementation process began with collecting data from the Spotify API. The data was then organized into a SQL database, cleaned and processed, and finally transferred into Python for machine learning. The final stage involved training and evaluating the models using standard performance metrics and comparing predictions with Spotify's actual popularity scores.

Part 1: API Setup & Data Collection

In this section I collected data from the Spotify API using my 'Spotify for Developers' account. After I created the account, I got a Client ID and Client secret that was unique to this project. Using the Client Credentials Flow I was able to obtain the public data such as track-level information (e.g., name, ID, release date), artist-level characteristics (e.g., popularity, genre tags). Then in Python I built a script that allowed me to obtain the Client Credential Flows and create a csv table. After downloading the proper python libraries, I created a script that allowed me to gather track-level and artist-level characteristics about songs from 2023. The script also created a dataframe for the output csv file.

track_id	track_name	album_name	release_date	explicit	track_pop
1k2pQc5i348DCHwn5KTdc	Pink Pony Club	The Rise and Fall of a Midwest Princess	9/22/2023	FALSE	
4KULAYmBBJCPRpk1yO4dOG	I Remember Everything (feat. Kacey Musgraves)	Zach Bryan	8/25/2023	FALSE	
51nyZpbJlJg2Iax3TlMamu	run for the hills	THINK LATER	12/8/2023	TRUE	
32aEs10B8C581q1YHpoQd6	I Smoked Away My Brain (f/m God x Demons Mashup) (feat. Imogen Heap & Clams Casino)	I Smoked Away My Brain (f/m God x Demons Mashup) (feat. Imogen Heap & Clams Casino)	8/18/2023	TRUE	
0eaVYo2zeOaGJeq25TWz	No Pole	Love Sick (Deluxe)	2/28/2023	TRUE	
3vkCueOmm7xQDdJ17W1Pm3	My Love Mine All Mine	The Land Is Inhospitable and So Are We	9/15/2023	FALSE	
7CyPwkp00E8Ro9Dd5CUDJW	One Of The Girls (with JENNIE, Lily Rose Depp)	The Idol Episode 4 (Music from the HBO Original Series)	6/23/2023	FALSE	
43lQbw5hx986dUEZbr3eN	From The Start	Bewitched	9/8/2023	FALSE	
36vmazy00iAE6FZT287fp2	Dive	Messy	6/30/2023	FALSE	
1TEBAk9toG0UJco6ZAUx	Rock and A Hard Place	Religiously. The Album.	1/11/2023	FALSE	
06BfR9yVSeuT4yrkuIn	Where the Wild Things Are	Gettin' Old	3/24/2023	FALSE	
3jd4wgM3cPeuLWMNHduBon	The Door	I've Tried Everything But Therapy (Part 1)	9/15/2023	FALSE	
4iuau93DlVte8XjntHdQJ	Brown Noise Therapy Loop For Sleeping (Loopable no fade)	Multicolored Noises For Sleep Or Focus (Loopable no fade)	8/1/2023	FALSE	
3Cp0iK69oQ1SdbB2J2ulGm	The Hardest Part	Messy	6/30/2023	FALSE	
0B97qFqGUnLwV2X2pXbf	Deep Sleep Brown Noise	Deep Sleep Brown Noise	6/2/2023	FALSE	
4S88neYCKT7yMlo2KOQEC	Thinkin' Bout Me	One Thing At A Time	3/3/2023	FALSE	
3rUGC1vUpjIDG9CZPHu1t	greedy	greedy	9/15/2023	TRUE	
1XBVIRV30ykhv54wm6En	NIGHTS LIKE THIS	THE FIRST TIME	11/10/2023	FALSE	
736PP5LTIREkDgkNmX3Gu	Superpowers	NEVER ENOUGH	4/7/2023	FALSE	
1Lo0QY9vc8sUB2vnlOXDt	Fast Car	Gettin' Old	3/24/2023	FALSE	
583dtdlUquLlMenarTp7PQm	Last Night	One Thing At A Time	3/3/2023	TRUE	
4mtmmBBEGdJLYWnLy5wWE	Cowgirls	One Thing At A Time	3/3/2023	FALSE	
3Gn1zuNHLuGGE0Xwq0k7e	Think I'm In Love With You	Higher	11/10/2023	FALSE	
17phzDn5oCzdlas6NuWj	Lose Control	I've Tried Everything But Therapy (Part 1)	9/15/2023	FALSE	
6393yGahSJ9sIVdQwSxOVR	Pink Pony Club	The Rise and Fall of a Midwest Princess	9/21/2023	FALSE	
98					

Changes after Part 1:

The initial proposal stated that I will be collecting track-level information, artist characteristics and audio features from the API. However, as I tried to code to get the audio features from the spotify API, I got a 403 error, that would not allow me to access the audio features. Upon doing research, I discovered that we cannot access Spotify's audio features anymore as it is not

public. However, this can be fixed by using a public static dataset that includes audio features such as tempo, energy, and danceability. The following adjustment should not change the direction and eventual outcome of this project.

Part 2: SQL & Data Integration

In this section I worked on integrating the static audio features dataset and building the SQL database for the project. Since Spotify no longer provides audio features through the API I downloaded a public dataset that contains information such as danceability energy tempo loudness acousticness valence and instrumentalness. After loading the dataset I cleaned the data by removing duplicates and standardizing the column names while also making sure the track IDs matched the API dataset from Part 1. Then in Python I created the SQL database by generating three tables for Tracks, Artists and AudioFeatures and imported both datasets into SQLite. Finally I wrote a join query that merged all three tables into one complete dataset which was exported as a CSV file for the machine learning stage of the project. After preparing the required libraries I wrote a script that cleaned the static audio features dataset and created the SQL tables for Tracks Artists and AudioFeatures. The script also loaded the data into the database and produced a merged view that combines all track artist and audio features into a single file ready for analysis.

CSV File Preview:

spotify_dataset																
track_id	artists	album_name	track_name	popularity	duration_ms	explicit	danceability	energy	key	loudness	mode	speechiness	acousticness	instrumentalness	liveness	valence
5SjWzRlFmHnGZGJgP7	Gen Hoshino	Comedy	Comedy	73	220560	FALSE	0.676	0.461	1	-6.745	0	0.143	0.0322	0.016	0.358	0.715
4PND8W1b13d_C09G3A	Ben Woodward	Ghost (Acoustic)	Ghost (Acoustic)	55	149610	FALSE	0.42	0.166	1	-17.235	1	0.0703	0.324	5.96E-06	0.101	0.267
1UB5v7N7yKA4MEG5G9Kb	Ingrid Michaelson,ZAYN	To Begin Again	To Begin Again	57	219260	FALSE	0.438	0.359	0	-9.734	1	0.0557	0.21	0.0	0.117	0.12
6lWq3G4KTEgTepGyQx	Kina Grannis	Orsay Rich Asians (Original Motion Picture Soundtrack)	Can't Help Falling In Love	71	201930	FALSE	0.266	0.0596	0	-18.515	1	0.0903	0.905	7.07E-05	0.132	0.143
5vJLS8fmP2RQ5SWu2K	Chord Overstreet	Hold On	Hold On	82	198830	FALSE	0.618	0.443	2	-9.681	1	0.0526	0.469	0.0	0.0829	0.167
01MV0BKvYNF8U874b	Tyrese Walls	Days I Will Remember	Days I Will Remember	58	214240	FALSE	0.688	0.481	6	-8.807	1	0.105	0.289	0.0	0.189	0.666
6V5wAMxK9GAM7Wu63tN	A Great Big World,Christina Aguilera	Is There Anybody Out There?	Say Something	74	229400	FALSE	0.407	0.147	2	-8.822	1	0.0355	0.857	2.89E-06	0.0913	0.0765
1EzEOckM9G43AKT7y7pA	Jason Mraz	We Sing, We Dance, We Steal Things.	I'm Yours	80	242940	FALSE	0.703	0.444	11	-9.331	1	0.0417	0.309	0.0	0.0973	0.712
0dK5u4GdG05G9MwZ8	Jason Mraz,Cobie Smollett	We Sing, We Dance, We Steal Things.	Lucky	74	199610	FALSE	0.625	0.414	0	-8.7	1	0.0399	0.294	0.0	0.151	0.889
7bK9u7uLdApqyE5kEsd	Ross Copperman	Hunger	Hunger	56	205594	FALSE	0.442	0.632	1	-8.77	1	0.0295	0.426	0.00419	0.0735	0.196
4mp5FnHkVGuShG5G4HTEJ	Zack Tabudlo	Episode	Give Me Your Forever	74	244800	FALSE	0.627	0.363	8	-10.127	1	0.0391	0.279	0.0	0.0928	0.301
5vF4eQb3uLV5AE9jPl	Jason Mraz	I Won't Give Up	I Won't Give Up	69	240165	FALSE	0.483	0.303	4	-10.058	1	0.0429	0.894	0.0	0.115	0.139
4pDuh3u5G7gQWwBwep	Dan Berk	Solo	Solo	52	198710	FALSE	0.489	0.314	7	-9.245	0	0.0331	0.749	0.0	0.113	0.807
0XMHxR17NehQjg5F2OO	Anna Hamilton	Bad Liar	Bad Liar	62	248448	FALSE	0.691	0.234	3	-6.441	1	0.0385	0.777	0.0	0.12	0.209
4LBWBN8Z2F9cgpb3	Chord Overstreet,Desp'nd	Hold On (Remix)	Hold On - Remix	56	188133	FALSE	0.755	0.78	2	-6.084	1	0.0327	0.124	2.83E-05	0.121	0.367
1K4b9R9R9Qwz9G9293MG	London Pegg	The Boy Who Never	Falling In Love at a Coffee Shop	58	244989	FALSE	0.489	0.561	4	-7.933	1	0.0274	0.2	4.56E-05	0.179	0.238
6uKcGy9vU3u3u1q4gm	Andrew Foy,Renee Foy	If I Love You Today	If I Love You Today	56	130750	FALSE	0.706	0.112	2	-18.998	1	0.0291	0.837	4.03E-06	0.125	0.414
47vGdpmoYvY1tscu4DOD	Andrew Foy,Renee Foy	At My Worst	At My Worst	54	169728	FALSE	0.795	0.0841	10	-18.09	0	0.0461	0.742	1.17E-05	0.0853	0.609
2qJLM8TEU3Cv9uLg4MMH8	Jason Mraz,Cobie Smollett	We Sing, We Dance, We Steal Things.	Lucky	68	199610	FALSE	0.625	0.414	0	-8.7	1	0.0399	0.294	0.0	0.151	0.889
6CgHuAFJ4G48tEJB8C	Boyce Avenue,Sea Miller	Cover Sessions, Vol. 4	Photograph	67	200186	FALSE	0.717	0.32	3	-8.393	1	0.0283	0.83	0.0	0.107	0.322
3B0CX2e9hdw4FY8WQz5uRW	Jason Mraz	We Sing, We Dance, We Steal Things.	I'm Yours	75	242940	FALSE	0.703	0.444	11	-9.331	1	0.0417	0.559	0.0	0.0973	0.712
21UJCvL5YD4YV8GZ9P	Boyce Avenue,Jennel Garcia	Cover Sessions, Vol. 3	Demons	63	174734	FALSE	0.678	0.351	0	-8.654	1	0.0266	0.747	0.0	0.355	0.699
2TE3p4G9F7GdSY9t4D	A Great Big World,Christina Aguilera	Is There Anybody Out There? - Track by Track Commentary	Say Something	70	229400	FALSE	0.407	0.147	2	-8.822	1	0.0355	0.857	2.89E-06	0.0913	0.0765
1K4b9R9R9Qwz9G9293MG	London Pegg	Coffee Moment	10 Million Miles	0	218386	FALSE	0.572	0.454	3	-10.986	1	0.0258	0.477	1.37E-05	0.0974	0.315
3HdLhGQd9vG9C9u3u7b	Jason Mraz	Human - Best Adult Pop Tunes	Unwired	0	231395	FALSE	0.796	0.607	5	-4.831	0	0.0392	0.361	0.0	0.221	0.754
4D33vK9WNEG9v9vU7r	Jason Mraz	Mellow Adult Pop	Bella Luna	1	302345	FALSE	0.755	0.454	9	-9.609	0	0.0352	0.757	0.0	0.236	0.53
8RQZ2RQZ2R2S8u4wE4PQ	Jason Mraz	Holly Jolly Christmas	Winter Wonderland	0	131760	FALSE	0.62	0.309	5	-9.209	1	0.0495	0.788	0.0	0.146	0.664
05x9bKtP25u4wMBwMwM	Jason Mraz	Feeling Good - Adult Pop Favorites	If It Kills Me	0	237953	FALSE	0.633	0.429	4	-8.764	0	0.0381	0.644	0.0	0.132	0.52
5QAMZTMs9uLgRfK5Z2T2	Jason Mraz	Christmas Time	Winter Wonderland	0	131760	FALSE	0.62	0.309	5	-9.209	1	0.0495	0.788	0.0	0.146	0.664
2gE8E1ZwYvT3Y7Xm0h	Jason Mraz	Perfect Christmas Hits	Winter Wonderland	0	131760	FALSE	0.62	0.309	5	-9.209	1	0.0495	0.788	0.0	0.146	0.664
3E2V12Hv9G9vRvH9b3g	Jason Mraz	Merry Christmas	Winter Wonderland	0	131760	FALSE	0.62	0.309	5	-9.209	1	0.0495	0.788	0.0	0.146	0.664
3u4H9vHv4R2L2L9KJ9	Jason Mraz	Christmas Music - Holiday Hits	Winter Wonderland	0	131760	FALSE	0.62	0.309	5	-9.209	1	0.0495	0.788	0.0	0.146	0.664
1NGLC9R859v9v9v9v9v9v	Chord Overstreet	Christmas Country Songs 2022	All I Want For Christmas Is A Real Good Tan	0	241946	FALSE	0.593	0.456	6	-8.193	0	0.0288	0.366	0.0	0.0914	0.654
3HdLhGQd9vG9C9u3u7b	Brandi Carlile,Sam Smith	Human - Best Adult Pop Tunes	Party of One	0	295558	FALSE	0.296	0.206	0	-11.799	1	0.0412	0.782	0.000225	0.0959	0.202
0dMRM9F9uq1q7WvWv9v	Brandi Carlile,Sam Smith	Mellow Adult - Adult Pop Favorites	Party of One	0	295558	FALSE	0.296	0.206	0	-11.799	1	0.0412	0.782	0.000225	0.0959	0.202
6H06144Z9Y1OX02u4f5d	Brandi Carlile,Sam Smith	Mellow Bars R'n'B	Party of One	0	295558	FALSE	0.296	0.206	0	-11.799	1	0.0412	0.782	0.000225	0.0959	0.202
6EwQLDw49p9C9uLwQ9uH1	KT Tunstall	Chil Christmas Dinner	Lonely This Christmas	0	257493	FALSE	0.409	0.153	6	-10.74	0	0.0306	0.939	2.56E-05	0.108	0.18
3Vw08bG9JdL2W9uG9G	Brandi Carlile	rainy day indie	Throwing Good After Bad	0	247791	FALSE	0.501	0.0852	2	-12.831	1	0.054	0.963	1.58E-05	0.0915	0.316
1w8X852DEmBFamGv	Brandi Carlile	Coffee Moment	This Time Tomorrow	0	206257	FALSE	0.606	0.306	10	-10.031	1	0.0294	0.841	0.0	0.139	0.414
6u4H9vHv4R2L2L9KJ9	KT Tunstall	seasonalchristmas	Lonely This Christmas	0	207480	FALSE	0.409	0.153	6	-10.74	0	0.0306	0.939	2.56E-05	0.108	0.18
6p9b9v9v9v9v9v9v9v9v	Little Velder	Mega Hit Autumn/Fall 2022	The Haves	0	306794	FALSE	0.474	0.519	7	-9.291	1	0.0253	0.281	0.0	0.107	0.320
1WFW9v9v9v9v9v9v9v9v	Brandi Carlile	Mellow Adult Pop	When You're Wrong	0	269690	FALSE	0.568	0.301	10	-10.163	0	0.0373	0.796	3.37E-06	0.118	0.142
0e9F9dY5ZDWW9v9v9v9v9v	Brandi Carlile,Luckie	Country Car Hits	You and Me on the Rock	0	230098	FALSE	0.568	0.686	1	-6.635	1	0.033	0.15	1.81E-06	0.0881	0.725
1MUL9v10Y09MG9L47Yv	Brandi Carlile,Luckie	Country Road Songs	You and Me on the Rock	0	230098	FALSE	0.568	0.686	1	-6.635	1	0.033	0.15	1.81E-06	0.0881	0.725
6U39v9v9v9v9v9v9v9v9v	Brandi Carlile	Finest Country	Speak Your Mind (from the Netflix Series "We The People")	0	109343	FALSE	0.476	0.666	6	-3.438	1	0.0446	0.314	0.0	0.342	0.498
7ADY9v9v9v9v9v9v9v9v9v	Brandi Carlile,Luckie	Easy Country	You and Me on the Rock	0	230098	FALSE	0.568	0.686	1	-6.635	1	0.033	0.15	1.81E-06	0.0881	0.725
3RE9v9v9v9v9v9v9v9v9v	Brandi Carlile,Luckie	Cozy Country	You and Me on the Rock	0	230098	FALSE	0.568	0.686	1	-6.635	1	0.033	0.15	1.81E-06	0.0881	0.725
UJ4H9v9v9v9v9v9v9v9v9v	Brandi Carlile,Luckie	Good Times, Country	You and Me on the Rock	0	230098	FALSE	0.568	0.686	1	-6.635	1	0.033	0.15	1.81E-06	0.0881	0.725
27H9v9v9v9v9v9v9v9v9v	Brandi Carlile,Luckie	Leafback Country	You and Me on the Rock	0	230098	FALSE	0.568	0.686	1	-6.635	1	0.033	0.15	1.81E-06	0.0881	0.725
1GBN9v9v9v9v9v9v9v9v9v	Brandi Carlile,Luckie	Chillin' R - Mellow Day Country	You and Me on the Rock	0	230098	FALSE	0.568	0.686	1	-6.635	1	0.033	0.15	1.81E-06	0.0881	0.725
72Y9v9v9v9v9v9v9v9v9v	Highland Peak	Trampoline - Acoustic	Trampoline - Acoustic	46	213098	FALSE	0.596	0.2	0	-10.424	0	0.0205	0.91	0.000183	0.0884	0.308
31LmwMv9v9v9v9v9v9v9v9v	Motolino Hata	Documentary	透明な日々世界	61	232380	FALSE	0.373	0.914	0	-4.185	1	0.0665	0.076	0.0	0.669	0.56
2DhUdA9v9v9v9v9v9v9v9v9v	Andrew Belle	Black Bear	Pieces	60	241119	FALSE	0.494	0.652	6	-5.863	0	0.0314	0.0754	0.02056	0.125	0.298
5JdC9v9v9v9v9v9v9v9v9v	Ron Pope	The Bedroom Demos	A Drop in the Ocean	68	202029	FALSE	0.447	0.393	5	-8.65	1	0.038	0.785	0.0	0.28	0.364
2y7F9v9v9v9v9v9v9v9v9v	Adam Christopher	So Far Away - Acoustic	So Far Away - Acoustic	52	171540	FALSE	0.576	0.331	6	-9.399	0	0.0309	0.894	3.44E-05	0.129	0.407
6u4H9v9v9v9v9v9v9v9v9v	Andrew Belle	Black Bear	The Enemy	62	266960	FALSE	0.536	0.89	1	-9.222	1	0.0375	0.078	0.0002	0.131	0.275
7L9v9v9v9v9v9v9v9v9v9v	Arns Weigt	Built It Better	Built It Better	51	234475	FALSE	0.526	0.333	4	-13.02	1	0.0314	0.91	0.198	0.111	0.159
25U9v9v9v9v9v9v9v9v9v9v	Chord Overstreet	Sleepwalking in the Rain	Sleepwalking in the Rain	0	216000	FALSE	0.501	0.381	0	-9.448	1	0.036	0.867	0.00133	0.119	0.302

Part 3: ML Learning

The final stage of the project focused on building machine learning models to predict Spotify popularity scores. The goal was to model the relationship between a track’s audio features, artist characteristics, and popularity score. The preprocessing stage involved loading the dataset, handling missing values, and performing feature engineering. Because music popularity is influenced by complex nonlinear relationships, Random Forest Regression was selected instead of relying only on simple linear regression. This ensemble method provided a stronger predictive baseline and better captured interactions between variables. The model was evaluated using standard regression metrics including Root Mean Squared Error (RMSE) and R-squared (R^2).

DataFrame (X and y)

Column	Type	Origin	Purpose in ML		
popularity	Int	Original	TARGET VARIABLE (y vector)		
danceability	Float64	Original	Numerical Feature		
energy	Float64	Original	Numerical Feature		
loudness	Float64	Original	Numerical Feature		
speechiness	Float64	Original	Numerical Feature		
acousticness	Float64	Original	Numerical Feature		
instrumentalness	Float64	Original	Numerical Feature		
liveness	Float64	Original	Numerical Feature		
valence	Float64	Original	Numerical Feature		
tempo	Float64	Original	Numerical Feature		
duration	Int	Original	Numerical Feature		
artist_popularity	Float64	Original	Numerical Feature		
release_year	Int	Engineered	Feature derived from release_date		
release_month	Int	Engineered	Feature derived from release_date		
key	Int	Original	Categorical Feature (Treated as numerical)		
mode	Int	Original	Categorical Feature (Treated as numerical)		
time_signature	Int	Original	Categorical Feature (Treated as numerical)		

Key Results & Findings

The Random Forest Regressor achieved strong predictive performance on the final 2023 Spotify dataset with an R^2 score of 0.852 and an RMSE value of 10.36. These results demonstrated that audio features such as danceability, energy, loudness, and tempo, combined with artist-level characteristics, explain a large portion of a song's popularity. At the same time, there is still a noticeable amount of variance that the model cannot explain. This is expected because many real-world factors influencing popularity are not included in the dataset, such as playlist placement, marketing strategies, social media trends, algorithmic recommendations, and Spotify's private engagement weighting system. Feature importance analysis showed that artist popularity was the strongest predictor. Other major contributors included danceability, energy, loudness, and tempo. These findings align with current streaming trends where energetic songs released by well-known artists tend to perform better on music platforms.

Advantages & Limitations

This project had several strengths and limitations. One major advantage was successfully creating a Spotify API workflow to retrieve metadata including track-level and artist-level information from 2023. Another advantage was integrating a Kaggle dataset containing audio features such as danceability, energy, and tempo. The combined datasets were cleaned and transformed into a structured SQL database consisting of multiple related tables. Additionally, Random Forest Regression effectively modeled interactions between different audio and artist characteristics. However, one of the largest limitations was Spotify removing public access to audio features through the API. Some tracks in the Spotify dataset also did not perfectly match the static Kaggle dataset, which may have reduced accuracy. Another limitation is that Spotify does not publicly disclose the exact formula behind its popularity metric. Finally, genre labels in the API were often inconsistent, making them difficult to use as standardized numerical features.

Conclusion

Overall, this project successfully built a system capable of predicting Spotify track popularity using real-world data management and machine learning techniques. Data was collected from both the Spotify API and Kaggle datasets, cleaned using pandas, and organized into a relational SQL database that served as the foundation for machine learning analysis. The final model not only predicted Spotify popularity scores effectively, but also revealed important relationships between audio features and artist-level characteristics. In the future, this project could be expanded by using larger datasets, incorporating additional variables, and experimenting with more advanced machine learning models to further improve predictive performance and better understand the factors that influence music popularity.

